

# Customer Segmentation & Prediction

## Business Recommendations

### Project Overview

The analysis segmented 500 customers into five customer groups using K-Means clustering and developed segment-specific Random Forest prediction models for churn analysis.

| Customer Segment                       | Observed Churn | Business Recommendations                                                                                      |
|----------------------------------------|----------------|---------------------------------------------------------------------------------------------------------------|
| Long-Term Customers                    | 7.22%          | Maintain loyalty programs, encourage renewals, and provide personalized account support.                      |
| High-Charge Electronic Check Customers | 10.86%         | Monitor high-value customers, provide targeted retention offers, and promote convenient payment alternatives. |
| Low-Churn One-Year Customers           | 3.23%          | Maintain customer experience, encourage renewals, and develop loyalty and referral programs.                  |
| Month-to-Month At-Risk Customers       | 20.59%         | Use targeted retention campaigns, communicate the benefits of longer contracts, and monitor churn closely.    |
| One-Year Electronic Check Customers    | 10.56%         | Encourage convenient payment alternatives and maintain proactive contract-renewal communication.              |

### Model Findings

- K-Means produced five customer segments used for the main prediction pipeline.
- Hierarchical Clustering was also evaluated as a second clustering method.
- DBSCAN was evaluated as a density-based clustering method and produced 35 clusters with 6.4% of observations classified as noise at the selected parameters.
- Random Forest models were trained separately for each customer segment.
- GridSearchCV was applied for hyperparameter tuning.
- Tenure was the most important feature across the segment-specific Random Forest models, followed by MonthlyCharges and TotalCharges.

### Model Limitations

- Some segments contain very few churned customers, which makes precision, recall, and F1-score less stable.
- The PCA visualization represents only 34.10% of total variance using its first two components.
- Feature importance indicates how the Random Forest models used the features; it does not establish causal relationships.
- Business recommendations should be validated with additional customer and business data before implementation.